THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS TWO-CHAPTER TEST SET D

Chapters 9 & 10: **The Baudhayana–Pythagoras Theorem, Proportional Reasoning-2** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The diagonal of a square of side 12 cm is: 1

(a) 24 cm **(b)** 12 cm **(c)** $12\sqrt{2}$ cm **(d)** $12\sqrt{3}$ cm

2. A Pythagorean triplet is: 1

(a) three consecutive numbers **(b)** three numbers with $a + b = c$

(c) three prime numbers **(d)** three numbers satisfying $a^2 + b^2 = c^2$

3. The triplet obtained from 2, 1 using $(m^2 - n^2,\ 2mn,\ m^2 + n^2)$ is: 1

(a) (1, 2, 3) **(b)** (3, 4, 5) **(c)** (4, 5, 6) **(d)** (2, 3, 5)

4. The diagonal of a rectangle 8 cm by 15 cm is: 1

(a) 23 cm **(b)** 13 cm **(c)** 19 cm **(d)** 17 cm

5. If the diagonal of a square is 10 cm, its side is: 1

(a) 10 cm **(b)** $5\sqrt{2}$ cm **(c)** 5 cm **(d)** 20 cm

6. A map scale 1 : 100000 means 1 cm on the map is how many km on the ground? 1

(a) 10 km **(b)** 1 km **(c)** 0.1 km **(d)** 100 km

7. If A : B = 2 : 3 and B : C = 3 : 5, then A : B : C is: 1

(a) 3 : 2 : 5 **(b)** 2 : 3 : 5 **(c)** 2 : 5 : 3 **(d)** 6 : 9 : 15

8. A scale drawing has scale 1 : 50. A 2 m wall is drawn with length: 1

(a) 40 cm **(b)** 10 cm **(c)** 4 cm **(d)** 2 cm

9. Three partners invest ₹ 2000, ₹ 3000 and ₹ 5000. The profit ratio is: 1

(a) 2 : 3 : 5 **(b)** 2 : 5 : 3 **(c)** 1 : 2 : 3 **(d)** 5 : 3 : 2

10. If a pizza is divided in the ratio 1 : 2 : 3 among three, the largest share is: 1

(a) 2/3 **(b)** 1/2 **(c)** 1/3 **(d)** 1/6

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Find the length of the diagonal of a cube face of edge 5 cm. 2

12. What is the side of a square whose diagonal is 14 cm? 2

13. A pie chart shows 300 students; 25% prefer cricket. How many? 2

14. A sum of ₹ 900 is shared in the ratio 4 : 5 : 6. Find the smallest share. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. A man walks 12 m north and then 5 m east. How far is he from the start? 3

16. A and B can do a job in 12 days and 15 days respectively. Who is faster and what is the ratio of their daily work? 3

17. Find the length of the diagonal of a rectangle whose perimeter is 34 cm and one side is 8 cm. 3

OR

Show that (2, 3, 4) is not a Pythagorean triplet while (9, 12, 15) is. 3

18. The ratio of the angles of a triangle is 2 : 3 : 4. Find each angle. 3

OR

The ratio of a father's age to son's is 7 : 2. If the difference is 35 years, find their ages. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. (i) Find the radius of a circle if a chord 24 cm is 5 cm from the centre.
(ii) Explain. 5

OR

(i) State the converse of Pythagoras theorem and give an application. 5
(ii) Check if 16, 63, 65 is a triplet.

20. (i) Show that if A : B = 3 : 4 and B : C = 5 : 6, then A : B : C = 15 : 20 : 24. 5
(ii) Find each if sum is 118.

OR

Case study: A scale on a floor plan is 1 cm : 2 m. A room is drawn 5 cm × 3.5 cm. 5
(i) Find the actual dimensions.
(ii) Area.
(iii) Cost of flooring at ₹ 450 per m².

END OF QUESTION PAPER — Total: 40 Marks

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